

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

ATTACHMENT F:

Transcript of Investigative Interview

(33 pages)

UNITED STATES OF AMERICA
 NATIONAL TRANSPORTATION SAFETY BOARD
 OFFICE OF ADMINISTRATIVE LAW JUDGES

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Investigation of:

BIG DIG CEILING COLLAPSE
 BOSTON, MA
 JULY 10, 2006

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* Docket No.: HWY-06-MH-024

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Interview of: FERNANDO CARVALHO

Offices of Nixon Peabody LLP
 100 Summer Street
 Boston, Massachusetts 02110

Wednesday,
 December 13, 2006

The above-captioned matter convened, pursuant to notice,
 at 11:30 a.m.

BEFORE: DENNIS COLLINS

APPEARANCES:

DENNIS COLLINS
National Transportation Safety Board

DAVID A. VICINANZO, ESQ.
LAURA E. COLTIN, ESQ.
Offices of Nixon Peabody LLP
100 Summer Street
Boston, Massachusetts 02110
(617) 345-1177

I N D E XITEMPAGE

Interview of Fernando Carvalho, Labor Foreman:

By Mr. Collins

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UNIDENTIFIED SPEAKER: Is it on yet?

MR. COLLINS: I'm going to find out. I'm going to see if it's working right now. So, ready? The first thing I'm going to do is identify myself for the transcriptionist. This is Dennis Collins with the National Transportation Safety Board.

Today is December 13, 2006. It's about 11:30 a.m., and I'm conducting an interview for the Big Dig investigation.

Sir, can you identify yourself, please?

MR. CARVALHO: Fernando Carvalho. I work for Mondern Continental (ph.), 23 years, almost 24, and I'm a labor foreman at the moment.

MR. COLLINS: Could you spell your name for me and for the transcriptionist?

MR. CARVALHO: First and last?

MR. COLLINS: Please.

MR. CARVALHO: F E R N A N D O, then I got a C in the middle -- C and C A R V A R -- no, no R -- L H O.

MR. COLLINS: Thank you. You said you had worked for Monarch for 23 years?

MR. CARVALHO: Yeah.

MR. COLLINS: Okay. Could you identify yourself, please?

MR. VICINANZO: Yes. Here as counsel to Fernando

1 Carvalho, I am David Vicinanzo, V as in Victor, I C I N A N Z O.

2 Also here is Laura Coltin.

3 MS. COLTIN: Laura Coltin, C O L T I N. I work with

4 David Vicinanzo.

5 MR. COLLINS: And I'm going to spell my name. This is

6 Dennis Collins again, and I'm going to spell my name for the

7 transcriptionist, Dennis, D E N N I S, Collins, C O L L I N S.

8 INTERVIEW OF FERNANDO CARVALHO

9 BY MR. COLLINS:

10 Q. Can I call you Fernando?

11 A. Yes. That's what everybody --

12 Q. Just wanted to make sure. All right. Fernando, could
13 you give me your date of birth and your address, please?

14 MR. VICINANZO: Is -- will this, I assume, be public?
15 He may not want his address out there.

16 MR. COLLINS: Well, the transcript will be public, but
17 before it's released, date of birth, any home addresses or phone
18 numbers and any other personally identifying information, we
19 redact.

20 MR. VICINANZO: You redact? Good, okay. All right.

21 MR. CARVALHO: -----.

22 MR. COLLINS: Okay.

23 MR. CARVALHO: And what else?

24 BY MR. COLLINS:

25 Q. Your address?

1 A. -----

2 Q. Could you spell that for me?

3 A. I have no idea. Let me look that up to make sure I

4 don't -----

5 Q. Okay.

6 A. -----

7 Q. Okay, and the city?

8 A. -----

9 Q. -----

10 A. Yeah.

11 Q. Okay. And the zip code?

12 A. -----

13 Q. Thank you.

14 MR. COLLINS: And, David, is there a number that --
15 should I -- let me back that up. If I needed to contact Fernando
16 for anything else, just go through you?

17 MR. VICINANZO: Just give me a call at (617) 345-1177.

18 MR. COLLINS: Okay. Okay.

19 BY MR. COLLINS:

20 Q. Now, I need you to think back about six years, to when
21 you were working on the tunnel. And if you could just tell me,
22 were you a labor foreman at that time?

23 A. Yes.

24 Q. Could you describe your duties for me? What were you
25 responsible for?

1 A. The staff of about three guys I had at that time with
2 me, and that -- beginning of the job, and do whatever the supers
3 asked me to do.

4 Q. Okay. Typically, what would the supers ask you to do?

5 A. Whatever has to be done at that time.

6 Q. Okay.

7 A. One of the things was start drilling. Start drilling
8 for the anchors, and to be cleaned up, because we'd be moving
9 stuff around.

10 Q. Okay. Let's go ahead and talk about the drilling
11 process. I understand that you did a little bit of it -- or, you
12 were shown how to do a little bit of it?

13 A. Yes.

14 Q. Could you walk me through the drilling process?

15 A. I think just for the first day or two, or maybe three --
16 I'm not sure how many days.

17 Q. Okay.

18 A. And then, the holes -- the surveyors lay out where we
19 have to drill, and the first day, or the first two or three days,
20 we have the -- or the guy there from Hilti (ph.) showing us how to
21 use the drill --

22 Q. Okay.

23 A. -- because it's something new. And I tell them, and the
24 drill has to be perfect, straight with a mark that would be on the
25 ceiling. The drill holes were marked, and like I said, it has to

1 be perfect, because in the beginning, we had a type of drill, it
2 was kind of -- was like a suction machine, okay? You can stick
3 them to the ceiling --

4 Q. Okay.

5 A. -- you want to kind of suck right into the concrete, and
6 you have to kind of play with that a lot, because it was not
7 always perfect. You weren't having perfect in the hole. You
8 never start drilling that hole, without the thing being perfect,
9 the way it's supposed to be. And a week later, we had a different
10 set up. It was the same drill with a different set up to be able
11 to move the drill the way you want them. It was easy for us to
12 get them perfect with that drill.

13 Q. Okay. The gentlemen from -- or the person from Hilti
14 was there for how many days?

15 A. Two -- three, I don't remember. Because, it was not
16 much to show, how to put the drill on, the drill off, and how to
17 set them for the concrete, and there's not much to learn.

18 Q. When you had the new drill a week later --

19 A. Was not a new drill, was the same drill.

20 Q. The new set up -- I misspoke --

21 A. Yes.

22 Q. -- the new set up to allow you to move the bit a little
23 bit --

24 A. Okay.

25 Q. -- did someone come out and show you the operation of

1 that as well, or --

2 A. It was very simple, very simple work. The same --
3 basically, it was the same set up, only it was something with a
4 nut you could just release and move the drill the way you want
5 them, or say if you got them in the right position, you tighten
6 that up, you are staying, instead of taking the whole thing apart,
7 put it back on, taking apart. It would have been easier to do it
8 that way.

9 Q. Okay. Did you have any help lifting the drill up?

10 A. Always two guys.

11 Q. Always two guys?

12 A. Yeah, yeah. There had to be two guys.

13 Q. Okay. Any sort of drill stand or other device that was
14 on the scissors lift with you, or it was up against the ceiling,
15 because you were holding it?

16 A. No. That was stuck to the ceiling.

17 Q. Okay.

18 A. You had, like, a compressor. You will suck right to the
19 ceiling. It would hold by itself anyway, but you don't have
20 to --after they touch the ceiling, you don't have to hold at all.

21 Q. After it was vacuumed to the ceiling, you could walk
22 away?

23 A. You could walk away --

24 Q. Okay.

25 A. -- because with the compressor always running, only if

1 the power shuts off for some reason -- you lose power --
2 something, then it would fall. With that, you have so many
3 minutes, because if the gauge -- it tells you on the compressor
4 when it's losing power or when it's losing air. Then that's when
5 you know right away, you know, always the guy watching that, too.
6 But the reason two guys, you know, one keeps an eye on that all
7 the time the -- and one can -- you just hold it.

8 Q. Only two of you on the lift at a time?

9 A. Yes.

10 Q. Okay. Do you remember what type of bits you used?

11 A. Was Hilti bits.

12 Q. Hilti bits. Okay. And do you remember how deep the
13 holes were to be drilled?

14 A. The hole -- I think, it has to be five -- between five
15 and five and-a-half. It couldn't be less than five. It couldn't
16 be more than five and-a-half, or something like that.

17 Q. How did you know when you got in the right bit?

18 A. The drill itself, it had, like, a stop.

19 Q. Okay.

20 A. You would set up a spot to drill that week, you know,
21 and you put a five and a quarter comes to stop right there. And
22 also, we used to mark them on the drill bit, too. That way, we
23 can see both, like with black markers just marking the -- or a
24 black tape along the bit, whatever we can see, whatever we got --
25 we always had there.

1 Q. Okay. And once you drill the hole and brought the drill
2 down, what did you do next?

3 A. Break the core to see if it's deep enough -- we called
4 it the core -- I don't remember the name of that thing, we break
5 them with a screwdriver, you check the hole and make sure we're
6 deep enough --

7 Q. Okay.

8 A. -- and make sure it's not too deep. I know if it's not
9 too deep, because if the drill run properly had to stop itself
10 right there.

11 Q. Would you measure it at that point?

12 A. After we're done, we measure.

13 Q. Okay. How many -- with the scissor lift up, how many
14 holes could you reach at one time, roughly?

15 A. It was two different set ups. It was a place we would
16 take four; there was a place we would take two.

17 Q. Okay.

18 A. But place that takes four, we can drill those four right
19 altogether, the same place. If there's only two, it's only two at
20 a time you can do -- you have to move the lift.

21 Q. Okay. How long would it take you to drill a hole,
22 roughly? And I know it's six years ago, so your best guess would
23 be fine.

24 A. I don't know. It would -- it would take ten minutes, or
25 it take five minutes. I'm not sure.

1 Q. What did you do when you encountered rebar?

2 A. I remember in the beginning, I know we drill them
3 through.

4 Q. Uh-huh.

5 A. In the beginning, I know we drilled them through -- we
6 had to stop, then, I don't know, Bechtel, I think they give the
7 okay to start drilling. So, I have no -- I don't remember what
8 happened.

9 Q. Okay. Was the -- if you hit -- I know you could drill
10 through rebar. If you hit something else, like one of the
11 junction boxes or something else, it's my understanding there were
12 various things in the ceiling. If you hit one of those, what
13 would you have to do?

14 A. No, we never hit those, because --

15 Q. Yeah.

16 A. -- we can see the box, we can see the doors right in the
17 ceiling.

18 Q. So you -- initially, you might, if you hit rebar, you'd
19 have to stop as you recall, and Bechtel came out and said no, you
20 can just drill through them, and then you just kept going through
21 them in the rebar?

22 A. Yeah, something like that. I'm not sure what happened
23 anymore.

24 Q. Then you didn't have a problem hitting anything else in
25 the ceiling?

1 A. No, because you would be able to see the box and see
2 everything with it.

3 Q. Okay. But -- let's move forward a little bit. You
4 drilled some holes. What was the next step after you drilled
5 them, what was the next thing you did to those holes?

6 A. Next thing, when I'm done with the holes and all that,
7 the next thing is that we would clean them.

8 Q. How did you clean them?

9 A. In the beginning, in the H0V side. That's why we have
10 to do the mock up -- they call them the mock up.

11 Q. Right.

12 A. I remember in the beginning, we had a nylon brush, and
13 Hilti comes with -- like an air squeezer -- they call them
14 squeezers. Like, remove the air, and we started with that.

15 Q. Okay.

16 A. You know, but where we had to start, and after we did
17 the first test and the first bolts, we had our -- I don't know, I
18 mean, it -- you know, for three or four, how many that didn't pass
19 the test, we did something different. We looked around, we looked
20 at electric compressors. We got one, and we look
21 -- we start looking for wire brushes to clean the holes even
22 better. It seemed better after that.

23 Q. Okay. Did you ever actually put any epoxy in the holes,
24 you, personally?

25 A. No, not personally, no.

1 Q. Did you ever get any training, or anyone show you how to
2 do that?

3 A. Well, I heard that, but I was out for a week, but one of
4 them came from the company --

5 Q. Okay.

6 A. -- to show the guys. But they was telling me they
7 there, I don't know, for a day, for two, I don't know -- or three,
8 I don't know. Was that week I was out. And they showed them how
9 to pump it out to load that, then start filling up in the hole.

10 Q. But you never --

11 MR. COLLINS: Go ahead.

12 MR. VICINANZO: The records are that he was on vacation
13 the week that the manufacturer came and trained them.

14 MR. COLLINS: Okay, great.

15 BY MR. COLLINS:

16 Q. And so, you never saw the training, and you never had
17 occasion to do anything yourself?

18 A. No, because I couldn't just stay up there, because like
19 I said, we were doing something else, I was, you know, checked
20 with the guys, everybody all set. But basically, in the
21 beginning, the -- those two same guys get ready for months to
22 drill, same two guys all the time, drilling and epoxy and things.

23 Q. Okay. Were you involved -- say the crew that you had
24 working drilling the holes and using epoxy ran out of epoxy,
25 didn't have any more. Do you know where they would get more from?

1 A. They would come to me; I need more epoxy.

2 Q. Okay. And then what would you do?

3 A. I would go up to what we call -- well, it's not really
4 the main office, but the office for the Boston area --

5 Q. Okay.

6 A. -- I would go get some more epoxy out of the office.

7 Q. And could you describe where that epoxy was stored for
8 me?

9 A. It was right in the office. There was a room right
10 there. It was inside there.

11 Q. Okay.

12 A. I would go out there and say I need the key or
13 something. If it was locked, I need a key from someone.

14 Q. And it was a -- not a trailer, but an actual building?

15 A. Office building, yes.

16 Q. How much epoxy was on hand at one time?

17 A. I don't know.

18 Q. When you went in to get more, was it, you know, two or
19 three boxes were there, or was it a pallet?

20 A. I know sometimes -- no, it was not a pallet. But two or
21 three boxes, because a lot of the times, I had to keep drilling,
22 because I had to wait for the epoxy to come in --

23 Q. Okay.

24 A. -- We don't have them in stock.

25 Q. When you went -- say you went to the office, and there

1 was no epoxy, what did you do then?

2 A. We just keep drilling.

3 Q. And, would you place an order somehow?

4 A. Yes -- that's the office, yeah.

5 Q. Okay. You just tell them --

6 A. I just -- yeah, we're out; we need more epoxy.

7 Q. All right. How long would it usually take to get the
8 epoxy once you told them they were out?

9 A. I think two or three days.

10 Q. Okay. Were you ever aware of any problems with the
11 drilling, other than, say, hitting rebar or with the use of the
12 epoxy?

13 A. No.

14 Q. Okay. You mentioned the testing, when you did the mock
15 up, and some of them had failed the testing. Did you ever observe
16 any of the testing? Get a chance to watch any of it?

17 A. I did, but I did not stay there -- no, I watch them test
18 one hole or something -- one hole, then I had to move on. You
19 know, I wouldn't stay.

20 Q. In both the mock up area and elsewhere, if a bolt failed
21 during the testing --

22 A. Uh-huh.

23 Q. -- what would you do?

24 A. Well, if the bolt failed, we have to kind of take the
25 epoxy right out --

1 Q. Okay.

2 A. -- redrill them, clean the hole back again, clean them
3 and all that, put the bolt back and retest them to see if they
4 passed.

5 Q. All right. So, if the bolt failed during the testing,
6 someone would tell you --

7 A. Yeah. They would have pulled the bolt right out.

8 Q. You'd send -- they pull the bolt out, you'd send the two
9 guys back up --

10 A. Yeah.

11 Q. -- they would redrill the hole. Would they use the same
12 drill?

13 A. Not necessarily. You can use the hammer drill. If the
14 hole is already made, you cannot change the location, because the
15 hole, it has to be perfect -- or the anchors, they have to be
16 perfect --

17 Q. Right.

18 A. -- for the plates to fit right on. And it's not
19 necessary if the hole is already made. All he's doing is just
20 cleaning the epoxy right up.

21 Q. Okay. And then once the old epoxy has been removed,
22 reclean the hole with the air, with the brush?

23 A. Yes. And epoxy the whole thing again, wait for how many
24 days we have to wait, and retest the bolt again.

25 Q. Okay.

1 A. It's not necessary we test them while there's just one
2 bolt. When we have more bolts to test it, then we would come, but
3 the other ones would test that one. But they would mark the
4 ceilings right away, and they would pull the bolts right out.

5 Q. So when the testing people were there, if a bolt failed,
6 they wrote something on the ceiling immediately --

7 A. Yes.

8 Q. -- and pulled the bolt out?

9 A. Yes.

10 Q. Throw the bolt away?

11 A. Yes. We don't use the bolt anymore.

12 Q. Okay.

13 A. And also Bechtel would be right there -- every time they
14 do a test, they would be right there with those guys all the time.

15 Q. Okay.

16 A. That's one thing they did.

17 Q. I had heard that Bechtel was there a lot, particularly
18 during the testing.

19 A. Yeah, and there were always two or three guys, all the
20 time.

21 Q. Okay.

22 A. We had one that was watching us, but not much in the
23 beginning to watch anyway, it was just us drilling, basically.
24 That time the crew was there.

25 Q. Okay. Were you in the area of the tunnel when the

1 hangers were hung off the bolts at all?

2 A. I was working in the tunnel, yeah.

3 Q. Were you involved in that process at all?

4 A. No.

5 Q. Okay. Did you ever have occasion -- and I'm going to
6 draw a little picture -- say you had a series of holes under the
7 four plates --

8 A. Uh-huh.

9 Q. -- and this hole -- for whatever reason, the bolt
10 failed, would you treat that like any other hole, drill it out and
11 put it in, or did you ever have occasion to have to move the whole
12 plate and redrill all the holes?

13 A. Only time I remember we had to do that was that if, for
14 some reason, the drill, like, if we were drill this plate --

15 Q. Uh-huh.

16 A. -- if, for some reason, like, say they got the drill
17 stuck in the concrete in that area, it could have been -- when
18 they built this tunnel, they built them with sidewalks and -- and
19 all that --

20 Q. Uh-huh.

21 A. -- and for this machine to suck right to the concrete,
22 the concrete would have to be really smooth. If, for some
23 reason -- and that -- just an example, if this area, I can't get
24 the drill to suck to drill them, I had to move the whole thing. I
25 have to ask the surveyors, you have to move this set up, this

1 plate.

2 Q. Then the procedure would be, if you couldn't get a good
3 suction here for this hole --

4 A. Yes.

5 Q. -- the surveyors would come out --

6 A. Uh-huh.

7 Q. -- and then relocate the hole?

8 A. Yes.

9 Q. Okay. Do you recall -- and I know you weren't directly
10 involved in it, you had said. Do you recall watching anyone
11 torque any of the nuts onto the bolts?

12 A. No.

13 Q. And I want to go over the epoxy story one more time,
14 just to make sure I have it correct. Where would you store it in
15 the tunnel? Would you -- how much would you keep on hand in the
16 tunnel, and where would you store it?

17 A. Basically, we had an office there in the tunnel, too.

18 Q. Okay.

19 A. That's where the supers used to hang around, and
20 sometimes Bechtel people came to do whatever they had to do. And
21 basically -- I wouldn't know the days they're going to do the
22 epoxy. And we can do 20 anchors or whatever, I would bring that
23 over. Whatever they're going to use for that day, I would bring
24 it over. I would have put them inside that office, and they grab
25 them when they need them.

1 Q. Okay. But you don't --

2 A. They brought them out, but when they tell me, then I
3 have to go back to the -- what we call the main office -- it's not
4 really the main office, but the office for the job right here in
5 Boston. I have to go over there and go get another five or six
6 cases, whatever -- no, not cases, box.

7 Q. Okay. So, just to make sure I understand it, you'd know
8 that you were going to be doing some epoxy today --

9 A. Uh-huh.

10 Q. -- so you'd grab five or six tubes --

11 A. Tubes.

12 Q. -- and bring them to the supers' trailer --

13 A. Yes.

14 Q. -- and then when they were getting to start working,
15 they'd grab them?

16 A. They'd grab them, yes.

17 Q. If, in the course of -- you know, they didn't finish the
18 first day, and the second day, they start working and they say,
19 Fernando, we're out of epoxy, you'd go back to the office and
20 bring some more --

21 A. Bring some more --

22 Q. -- give them what they need --

23 A. Yes.

24 Q. -- and put some in the supervisor's trailer?

25 A. Uh-huh.

1 Q. Where was the super's trailer?

2 A. It was right inside the tunnel.

3 Q. Right inside the tunnel?

4 A. Yeah, whatever we had power to hook up to.

5 Q. Air conditioned and heated? What was the temperature
6 like inside the trailer?

7 A. Always warm, always in the 50s. In the winter, we had
8 -- because we're doing tile at the same time on the walls --

9 Q. Uh-huh.

10 A. -- we always have the ends to covered with tarps and
11 heaters, electric heaters inside the tunnel to keep the same
12 temperature all the time.

13 Q. Okay. So, the temperature in the trailer, in that part
14 of the tunnel, never went below --

15 A. Well, in the trailer, they had the heater on itself, and
16 we got electric heaters, too.

17 Q. What's that?

18 A. We had the trailer themselves, whatever was there, the
19 tiles, the guys are working and then whatever, we always had
20 heaters -- electric heaters.

21 Q. So, you would say it never got below --

22 A. It had to be above 45, if I'm not wrong, to do the tile.

23 Q. Okay.

24 A. It had to be above 45 -- or it couldn't be below 45, or
25 something like that. That's why you always have to -- the

1 temperature always has to be higher.

2 Q. Okay. Did the guys working on the lift ever tell you
3 they had any problems with the epoxy --

4 A. No.

5 Q. -- a bad bottle of --

6 A. No.

7 Q. Any problems with the gun or the nozzles?

8 A. No. We store the nozzles and gun inside the tunnel. If
9 the gun was not working, they would ask me for another gun,
10 because sometimes they break after a while.

11 Q. Do you know -- I know you weren't there for the
12 training, but in watching them work or talking with the laborers,
13 do you know how deep into the hole they went with the nozzle?

14 A. I think they went in as far as it could go, but it's not
15 that much. It's only five inches or five and-a-half, and they
16 start right from the top, working their way down.

17 Q. Okay.

18 A. And the holes themselves -- they have like the red cap
19 to hold the anchor.

20 Q. And you said you have 23 years experience -- was it 23
21 years then or 23 years now, so I can --

22 A. I started in '83.

23 Q. '83. Okay, that's fine. Do you remember anything out
24 of the ordinary or unusual about the drilling or the use of the
25 epoxy at the time, anything?

1 A. No.

2 Q. Okay. Did the fellows working with the epoxy, did they
3 wear any protective equipment?

4 A. No, not really. Regular clothes. Probably, they used
5 gloves, but I'm pretty sure, I think, rubber gloves that they were
6 using, but if the epoxy get on it, you have to take them out
7 again.

8 Q. I had occasion to work with a little bit of it in our
9 lab, and I certainly -- I bet after the first time you had some on
10 you, you'd wear gloves.

11 A. Now I'm thinking about it, yeah, they used rubber
12 gloves, I'm sure.

13 Q. Well, I think -- if you'd had any problems, say the one
14 time -- let me back that up. When a bolt fails, I'm sure that the
15 inspectors reported that failure to someone. If you had any
16 problems, for example, when you first hit rebar, who was the
17 person you talked to that said, hey, we're hitting rebar, what
18 should we do?

19 A. Was the supers.

20 Q. The supers?

21 A. Yes.

22 Q. Who were the supers, do you remember?

23 A. Bill Brito (ph.) and Scott Abbotts (ph.).

24 MR. VICINANZO: Abbott (ph.).

25 MR. COLLINS: Brito and --

1 MR. VICINANZO: Bill Brito and Scott Abbott.

2 MR. COLLINS: And Abbott. Okay.

3 BY MR. COLLINS:

4 Q. You never heard anything in the ceiling when you had to
5 move that one -- you said you had -- you remember once, maybe
6 twice --

7 A. Yeah.

8 Q. -- not being able to --

9 A. Because the ceiling was not as straight --

10 Q. So, you would do the same thing, go to one of the
11 supervisors and say -- and tell them --

12 A. Yeah. You have to send to surveyors, yes, and lay out
13 new hole.

14 Q. How long would it take the surveyors to come out and lay
15 a new hole?

16 A. Well, I put them back two plated behind and keep
17 going --

18 Q. Okay.

19 A. -- before I would not really have the men drill.

20 Q. You -- if I'm understanding you correctly, you never
21 stopped working. If you had a problem with one plate or one
22 hole --

23 A. We'd move onto the next one.

24 Q. -- you'd notify the next person --

25 A. Yes.

1 Q. -- of the change and kept going?

2 A. Yes.

3 Q. And at some point, you'd go back and do whatever to fix
4 it?

5 A. Yeah.

6 Q. Be it drill it out and put a new bolt in --

7 A. Yes.

8 Q. -- move the plate? Okay. When the surveyors marked the
9 holes -- locations, did they --

10 A. Used, like, a template.

11 Q. They did use a template?

12 A. Yes.

13 Q. Okay.

14 A. Made out of a plate, whatever that plate that was there,
15 made out of it. I had one, too, to fix the bolts.

16 Q. Uh-huh.

17 A. I had one of those templates, too, just like that --

18 Q. Okay.

19 A. -- to fix the bolts, to make sure the bolts were nice
20 and straight, to go in and out.

21 Q. So, once you'd place the --

22 A. And right at the same time, we're placing the bolts in,
23 the second guy -- because there would always be two guys, always
24 trying to make sure the bolt, you know, the template would go in
25 and out the way it's supposed to go.

1 Q. Okay. Let's say I'm working with one of these four-hole
2 plates. We -- when the holes are drilled, and they've been
3 cleaned, and they're dry, then I go to put the epoxy in, I put the
4 red caps in, put the epoxy in, one, two, three, four -- four
5 bolts --

6 A. We used to do a check, and even before we start with the
7 anchors, but we had to place a bolt --

8 Q. Oh, okay.

9 A. -- yeah, we'd put them right there in the hole. We'd
10 know if they line up.

11 Q. Holes are drilled and cleaned, we're going to put the
12 template out to make sure we're good?

13 A. Yes. But they've got to line up, but when we have to
14 cross that one, we move, we tell them to move.

15 Q. Okay.

16 A. That would -- it never happened. As far as I remember,
17 it never happened. I take -- we never had to move. But we check.

18 Q. Never had to move them, because they didn't line up?

19 A. Yes.

20 Q. But you checked every one?

21 A. Yes.

22 Q. Did you have -- you said there were some that were two
23 holes? Did you check those as well?

24 A. Yeah. Plates with four holes and plates with two holes.

25 Q. And you would have those handy to check the holes --

1 A. Yes.

2 Q. -- before you put the epoxy in?

3 A. It's a piece of plywood.

4 Q. Okay. I didn't know that. I learned that today. And
5 did you have -- you described the attachment to hold the drill to
6 the ceiling. It was vacuumed, and then it was the one you could
7 adjust -- did you ever have a problem with the drill bit skipping
8 when you started the holes? Or did the attachment help you keep
9 the bit right on target?

10 A. No, the bit -- they have to be right on target, and if
11 they're -- no, the bit wouldn't move.

12 Q. No? It wouldn't move?

13 A. No.

14 Q. With the core drill -- I'm trying to formulate my
15 question.

16 A. With a hammer drill, you could do that. You could
17 move -- that was the set up. The set up -- not the set up we had.
18 It was like a stand that the drill would travel, and the stand
19 like this.

20 Q. Okay.

21 A. It would come -- it can't move.

22 Q. Okay, once you line it up, the drill goes straight up --

23 A. Yes.

24 Q. -- and straight down?

25 A. Yes.

- 1 Q. Okay.
- 2 A. But it travels in the stand. It's a stand -- I'm not --
- 3 Q. Me neither. Did you see that it's clear?
- 4 A. It's a stand right here. That's what the -- just in the
- 5 couple of the trailers that was something like that. The drill
- 6 travels in that back and forth, up and back.
- 7 Q. Okay.
- 8 A. Travels in this, like --
- 9 Q. And this would be the ceiling of the tunnel?
- 10 A. Yes. It would stop right at that, and the drill would
- 11 kind of go --
- 12 Q. And you would vacuum attach that to the ceiling --
- 13 A. Uh-huh.
- 14 Q. -- so that the bit was on the mark?
- 15 A. Yes.
- 16 Q. And then, the drill would be allowed to move up and
- 17 down?
- 18 A. Up and down, but not --
- 19 Q. But not side to side?
- 20 A. No. No, like I said, then, two weeks after, or a week
- 21 after, I don't remember when we got the other set up, the whole
- 22 stand moves, not just the drill. It was the whole stand move to
- 23 line up with the hole where we -- to be right on that, instead of
- 24 taking the whole thing right out --
- 25 Q. Uh-huh.

1 A. -- tried to put them back on, then it was not level. We
2 bring the drill out. If it's not level, we have to take it down
3 again, try to set up again. The other way was, like, twice. We
4 used to do twice the work. And I was killing myself.

5 Q. Was that -- with the first system, you'd do twice the
6 work?

7 A. Yes.

8 Q. With the second one, you could make --

9 A. They'd do twice the work, yeah. Because when I was
10 killing myself to get fit, you have to keep going up and down
11 until they got them set. The other one, you'd set up one time,
12 right there --

13 Q. Uh-huh.

14 A. -- then they do what you have to do.

15 Q. I think that's it. I think that's all I needed, unless
16 you can think of anything else that would help me understand the
17 procedure for installing the bolts and the epoxy, and hanging the
18 panels, or anything else that might help us figure out why this
19 particular system did not do what it was supposed to do.

20 A. No. I don't know if it could be the traffic on the top
21 on that area, I don't know. I was working underneath there, and I
22 never thought that thing was going to come down.

23 Q. You didn't have any problems working under the panels,
24 or --

25 A. No, no.

1 Q. Perfectly comfortable?

2 A. Yes.

3 Q. Okay.

4 MR. COLLINS: Dave, do you have anything?

5 MR. VICINANZO: No, I don't have a thing.

6 (Whereupon, the interview in the above-entitled matter
7 was concluded.)

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CERTIFICATE

This is to certify that the attached proceeding before the

NATIONAL TRANSPORTATION SAFETY BOARD

IN THE MATTER OF: Big Dig Ceiling Collapse
 Boston, MA
 July 10, 2006

DOCKET NUMBER: HWY-06-MH-024

PLACE: Boston, Massachusetts

DATE: December 13, 2006

was held according to the record, and that this is the original,
complete, true and accurate transcript which has been compared to
the recording accomplished at the hearing.

Pamela J. Legg
Transcriber